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Theoretical calculation of keyhole aperture morphology in laser deep penetration welding

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Abstract

The keyhole, as the primary feature of laser deep penetration welding, represents the principal site for achieving light-thermal energy conversion in the welding process. To study the three-dimensional morphology of keyhole, it was assumed in this paper that the temperature of molten liquid at the edge of keyhole aperture is boiling point temperature, and the boiling temperature isotherm on melt pool surface was calculated based on two different treatment methods of heat source to estimate keyhole aperture profile. Then it was subsequently verified by using in-situ optical observation and the method about melt pool quick freezing to retain keyhole. The results show that keyhole aperture in laser deep penetration welding can be divided into the zone formed by laser direct heating and the zone formed by laser-induced vapor eruption from front keyhole

Outline

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wall. The former is closely related to the laser energy distribution on material surface rather than the laser energy distribution in the direction of keyhole depth, while the latter is more susceptible to the influence of erupted laser-induced vapor related to the energy distribution on front keyhole wall. When the welding speed is low, the tilt angle of front keyhole wall is larger, the zone formed by laser-induced vapor eruption from the front keyhole wall is not obvious, and the zone formed by laser direct heating is approximately circular in shape. The profile of boiling temperature on the laser-induced melting pool surface calculated by the direct calculation method of laser heat source acted on material surface is consistent with the zone formed by laser direct heating. When the inclination angle of front keyhole wall is reduced by changing welding speed or material, the impact of laser-induced vapor emitted from front wall on rear wall is increased, resulting in an elliptical or “gourd” morphology of the keyhole aperture.

Keywords

Laser deep penetration welding; Keyhole aperture; Profile estimation; In-situ observation; Evaporation vapor

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1. Introduction

Laser deep penetration welding technology is widely utilized in manufacturing fields including aerospace, automotive manufacturing, rail transportation, ships and naval vessels due to its advantages of high welding speed, high welding quality and high processing flexibility [1], [2], [3], [4]. The essence of laser deep penetration welding process is the utilization of laser thermal effect. The keyhole, as the signature characteristic of laser deep penetration welding process [5], [6], represents the primary site for conversion of laser energy into the heat energy of molten pool. The three-

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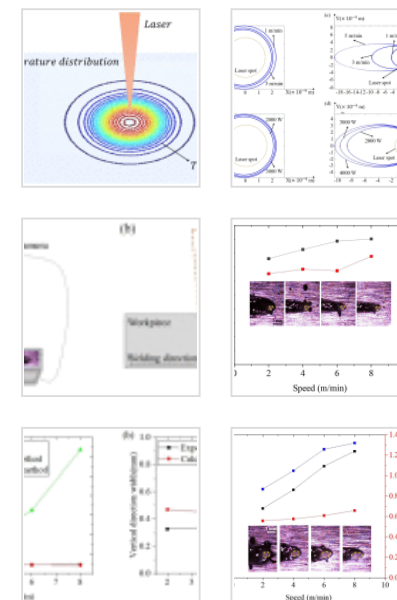
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dimensional morphology of keyhole is therefore one of the crucial factors in the comprehension of laser-matter interaction pattern in welding process. And the keyhole aperture morphology plays a pivotal role in the three-dimensional morphology of keyhole. Consequently, the study of the calculation method and the formation mechanism of keyhole aperture is vital for establishing three-dimensional keyhole morphology and comprehending physical process of laser deep penetration welding.

In recent years, scholars have used high-speed cameras [7], [8] and X-ray systems [9], [10] to observe the morphology of keyhole aperture in laser deep penetration welding directly, and the formation mechanism and influencing factors were analyzed. Pan et al. [11] used a high-speed camera to observe the keyhole aperture morphology of galvanized and ungalvanized steel with different thicknesses welded by Nd: YAG laser at different welding speed, and found that zinc vapor had a significant influence on the characteristics of keyhole aperture and the nearby molten pool. Aalderink et al. [12] used a high-speed camera to observe the keyhole aperture morphology during Nd: YAG laser welding of aluminum alloys and concluded that the change of keyhole pressure balance was responsible for the evolution of keyhole aperture morphology from round to oval. Sato et al. [13] used X-ray imaging technology to observe the keyhole aperture morphology during disc laser welding of 304 stainless steel at different ambient pressures and found that the lower the ambient pressure, the greater the keyhole aperture width and concluded that the keyhole depth depends on the size of keyhole aperture.

Due to the presence of metal vapor and raised melt in the vicinity of keyhole aperture during laser deep penetration welding, the visual observations are mostly based on image processing methods to obtain a clearer outline of keyhole aperture and establish further relationships with welding process. Luo et al. [14] observed the geometry of the top keyhole aperture during fiber laser welding of 304 stainless steel using a coaxial monitoring system, and established a neural network model between the geometry modified by binarization or other image processing methods and the porosity to achieve the prediction of welding defects. Zhao et al. [15] established a coaxial visual monitoring

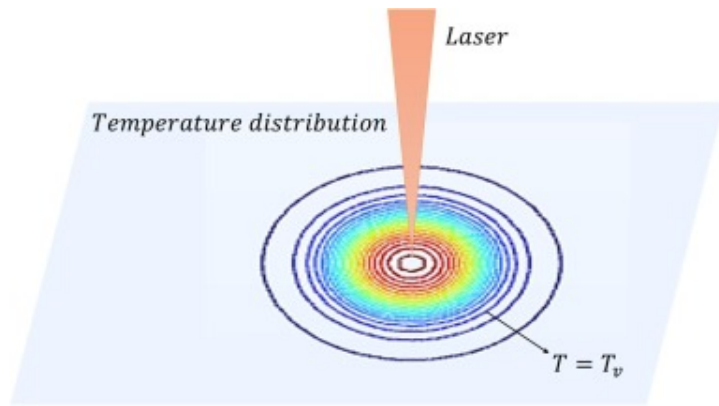
system with a laser-assisted light source, and proposed a hybrid adaptive keyhole detection algorithm to reduce the interference of specular reflections and metal vapor on the photographic results of keyhole aperture during the fiber laser welding of stainless steel, and the relationship between keyhole characteristics and weld penetration state was established. Zhang et al. [16] used the area growth algorithm and the Canny algorithm to process the keyhole aperture profile from high-speed camera photography during fiber laser welding of galvanized steel, enabling online monitoring of weld quality during laser deep penetration welding by establishing the relationship between keyhole aperture and melt pool depth in different penetration states.

In summary, the study of keyhole aperture contour, the formation mechanism and its influencing factors can reveal the energy coupling law in keyhole and provide a basis for the calculation of keyhole depth. Additionally, it can provide theoretical support for the welding process quality monitoring based on keyhole aperture in situ observation. In recent years, the direct calculation of keyhole aperture profile in domestic and foreign studies is relatively few, and the partial characteristics of keyhole aperture can be mostly concluded in some calculations for keyhole. For example, Dowden et al. [17] assumed that the keyhole was circular and the keyhole aperture width was solved as a variable in the relevant implicit function for describing keyhole. Kaplan et al. [18] calculated the geometry of an irregular keyhole wall parallel to welding direction point by point by solving the local energy balance at any position on keyhole wall, thus the keyhole aperture width along the welding direction can be obtained. However, most studies ignored the differences of keyhole aperture characteristics under different parameters. And it was more directly assumed that the keyhole aperture morphology is similar to the focus spot on the material surface [19], [20]. Alternatively, it is assumed that the keyhole aperture radius is equal to or slightly larger than the spot radius, or the keyhole aperture is oval [21]. At the same time, most research focused on providing a recognizable keyhole aperture feature for process monitoring through welding experiments. However, there are few discussions on the physical mechanism of keyhole aperture itself through the combination of heat-transfer theoretical calculation and experiments.

In this paper, we proposed a method to characterize the keyhole aperture during laser deep penetration welding by conducting the boiling temperature boundary on the surface of a laser-induced molten pool, and compared the two methods of estimating the boiling temperature isotherm of melt pool surface based on heat transfer laws. At the same time, the keyhole aperture profile was observed in situ by a high-speed camera during fiber laser welding of low-carbon steel and the keyhole was retained based on the method of melt pool quick freezing during the welding of copper-hafnium alloy. Subsequently, the estimation method of keyhole aperture profile was verified. Afterward, based on the comparison results of theoretical calculation and experimental measurement, the reasons for the difference between calculated and experimental results and the influence law of laser-induced vapor from the keyhole wall on keyhole aperture morphology were analyzed and discussed.

2. Calculation methods of the keyhole aperture profile

In this paper, it is assumed that the region where the surface temperature of laser-induced molten pool reaches the boiling temperature of material T_v is the range of keyhole aperture. Consequently, the keyhole aperture profile is the temperature isotherm at T_v on molten pool surface, as shown in Fig. 1. In the calculating process of keyhole aperture profile (that is, calculating the temperature isotherm at T_v on molten pool surface), only the laser energy absorption of molten pool surface is considered. The absorption coefficient A is assumed to be constant. The influence caused by the evaporating vapor from keyhole wall and the influence of plume on laser beam transmission are ignored. On this basis, the energy distribution of incident laser I and the energy required for material to reach the boiling temperature q_v must be firstly determined in the calculation of temperature isotherm at T_v on molten pool surface.



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Fig. 1. Schematic diagram of the boiling temperature isotherm at the keyhole aperture.

2.1. Energy distribution of the incident laser

During the calculation, the laser energy distribution is assumed to follow Gaussian distribution, which can be expressed as:

$$I(\mathbf{r}) = \frac{2P}{\pi\delta(z)^2} \exp\left(-\frac{2r^2}{\delta(z)^2}\right) \quad (1)$$

where P represents the laser power. $\mathbf{r}$ represents the distance from spot center. According to the propagation principle of Gaussian beam, the laser spot radius at different depth positions $\delta(z)$ can be expressed as:

$$\delta(z)^2 = \delta(z_0)^2 + \alpha^2(z - z_0)^2 \quad (2)$$

where $\delta(z)$ represents the laser spot radius at the depth z , $\delta(z_0)$ represents the laser spot radius at the focal point, z_0 represents the focal plane coordinates, and the beam focus angle (half angle) can be expressed as:

$$\alpha = \frac{D}{2f} \quad (3)$$

where D represents the diameter of laser spot before focusing, and f represents the focal length of focusing lens, and the radius of focused spot at the focal plane position can be expressed as:

$$\delta(z_0) = \frac{2K_f \cdot f}{D} \quad (4)$$

where K_f represents the value of beam-focusing characteristic parameter. From Eqs. (2)-(4), the laser spot radius at different depth positions can be expressed as:

$$\delta(z) = \sqrt{\frac{4K_f^2 \cdot f^2}{D^2} + \frac{D^2}{4f^2} \cdot (z + \Delta F)^2} \quad (5)$$

where ΔF represents the defocus amount. From Eqs. (1)-(5), the laser beam energy distribution can be obtained for different focusing systems (focusing mirror focal length, beam focus angle) and beam quality (K_f), etc.

2.2. Energy required for the material to reach boiling temperature

In this paper, it is considered that the energy q_v required for material to reach boiling temperature under laser action can be divided into two parts. The first part is the energy required to raise the material temperature from ambient temperature to boiling temperature considering only the heat conduction; the second part is the energy absorbed by the material to overcome its latent heat of fusion.

The heat flow from keyhole wall into the workpiece can be resolved by the moving line heat source model proposed by Rosenthal et al. [22], which gives the temperature field distribution at a point in the workpiece:

$$T(\mathbf{r}, \varphi) = T_a + \frac{P'}{2\pi\lambda} K_0 (Pe' r) e^{-Pe' r \cos \varphi} \quad (6)$$

where $(\mathbf{r}, \varphi)$ represents the position coordinates with the line source as the origin; $Pe' = v/2\kappa$ represents the Peclet number; v represents the welding speed; $\kappa = \lambda/\rho c$ represents the thermal diffusivity of material; λ represents the thermal conductivity; ρ

represents the material density; c represents the heat capacity; T_a represents the ambient temperature; P' represents the intensity of line source; $K_0()$ represents the modified Bessel function of second kind and zeroth order. The heat source intensity required to reach the boiling temperature at any point (r, φ) on keyhole wall can be expressed as:

$$P'(r, \varphi) = (T_v - T_a) 2\pi\lambda \frac{1}{K_0(Pe'r)} e^{Pe'r \cos\varphi} \quad (7)$$

Combined with Fourier's law of heat conduction, it is simplified by considering only the radial component. The heat flow required to reach boiling temperature at a point on keyhole wall can be expressed as:

$$q_{v1}(r, \varphi) = (T_v - T_0) \lambda P e' \left(\cos\varphi + \frac{K_1(Pe'r)}{K_0(Pe'r)} \right) \quad (8)$$

where $K_1()$ represents the modified Bessel function of the second kind and first order.

The other part of the energy required to reach boiling temperature is the energy absorbed during the material melting to overcome its latent heat of fusion, which can be expressed as [23]:

$$q_{v2} = \rho_s L_m \bullet v \quad (9)$$

where ρ_s represents the density of solid material; L_m represents the latent heat of fusion; v represents the welding speed. By combining Eqs. (8)-(9), the energy required for the material on keyhole wall to reach boiling temperature can be expressed as:

$$q_v(r, \varphi) = (T_v - T_0) \lambda P e' \left(\cos\varphi + \frac{K_1(Pe'r)}{K_0(Pe'r)} \right) + \rho_s L_m \bullet v \quad (10)$$

2.3. Calculation of the isotherm at T_v of the molten pool surface temperature by laser action

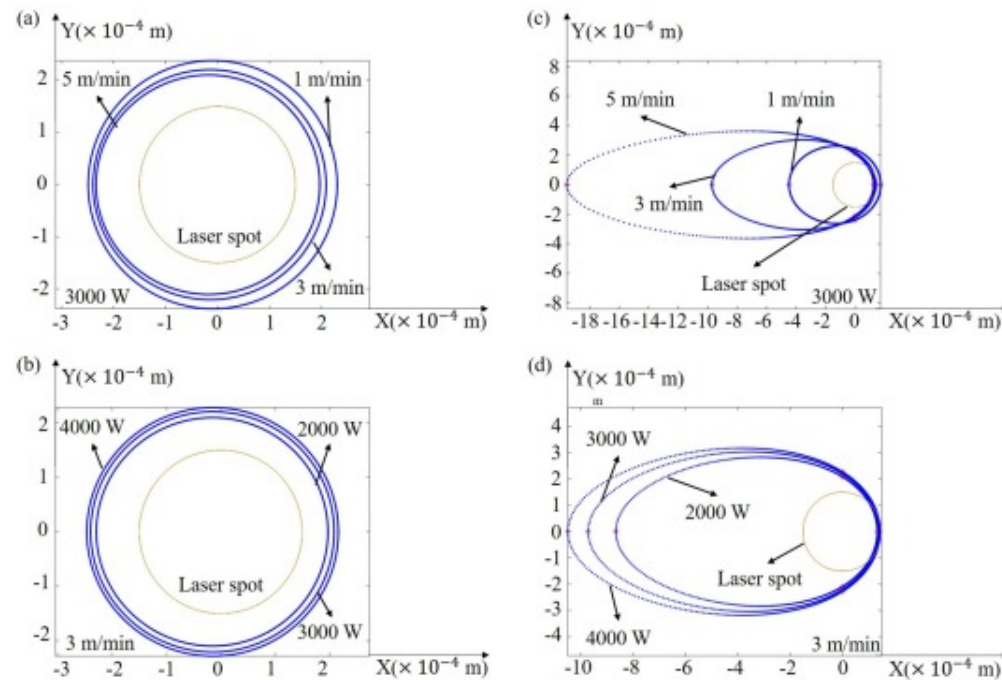
In this paper, two methods were used to calculate the temperature isotherm at T_v on molten pool surface based on the law of heat conduction. The first method is the direct calculation about the material surface heating by laser heat source (hereinafter referred

to as the direct calculation method); the second method is the indirect calculation using the point heat source conducted by the energy redistribution on material surface (hereinafter referred to as the indirect calculation method). The direct calculation method is to disregard whether the keyhole is formed or not. Given the laser power density distribution I on material surface, combined with the energy q_v required for the material to reach boiling temperature at different positions $(\mathbf{r}, \varphi)$, the isotherm of boiling temperature on molten pool surface, which is the morphology of keyhole aperture profile assumed in this paper, can be obtained by finding the set of all position coordinates $\mathbf{r}$ satisfying the critical condition about $q_v \leq A \bullet I$ under the different values about φ by setting $0 < \varphi < 2\pi$.

The indirect calculation method assumes that the keyhole has been formed. Then the heat source can be equated to a combination of in-plane point heat sources at different depth locations, and there are differences in the intensity of the in-plane point heat sources. Therefore, an energy redistribution calculation is performed to determine the intensity of the in-plane point heat source on material surface. Firstly, the keyhole aperture width corresponding perpendicular to welding velocity direction when $\varphi = \pi/2$ is calculated. Based on this, the intensity of the point heat source P' on material surface is determined according to Eq. (7). Afterward, the temperature field distribution of this heat source on material surface based on heat conduction is calculated according to Eq. (6). The isotherm of boiling temperature on molten pool surface is calculated by setting $T(\mathbf{r}, \varphi) = T_v$. Then, the keyhole aperture profile morphology and the keyhole aperture width in the direction parallel and perpendicular to welding speed can be obtained.

For the calculations in this paper, a fiber laser was used to weld thick plates of low-carbon steel material. The radius of laser focus spot on the material surface was approximately 0.15mm, the laser power was 3kW, and the welding speed was set to 1, 3 and 5m/min. The profile morphology of keyhole aperture was obtained by using MATLAB software based on the two different algorithms, as shown in Fig. 2. After that, the welding speed was fixed at 3m/min and the laser power was changed to 2, 3 and 4kW. The profile morphology of the keyhole aperture corresponding to the two different algorithms was

obtained respectively, as shown in Fig. 2. As can be seen in Fig. 2, the keyhole aperture profile obtained by the direct calculation method is approximately circular, with the center of laser spot slightly deviating from keyhole center along the welding direction. But the indirect method gives an elliptical keyhole aperture profile and the laser spot is mainly located in front of the keyhole aperture. Meanwhile, for the width of keyhole aperture in the parallel and perpendicular direction to welding speed, the results calculated by the direct calculation method decrease with increasing welding speed and decreasing laser power. The results calculated by the indirect calculation method correspond to the same pattern of variation with laser power as the first algorithm and the opposite pattern of variation with welding speed.



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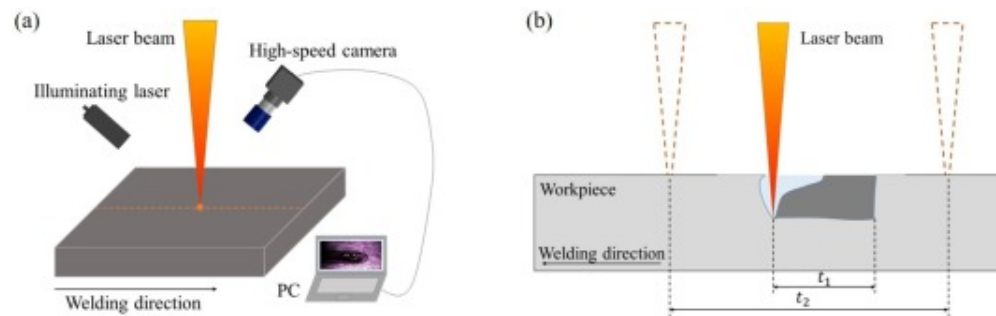
Fig. 2. The relationship between the keyhole profile calculated by the direct calculation method (a), (b) and the indirect calculation method (c), (d) with the variation of welding

speed and laser power respectively.

3. Experimental validation of the method for calculating the keyhole aperture profile morphology

3.1. Experimental setup and methods

An IPG YLS-6000 fiber laser with a rated output of 6kW and a beam quality parameter K_f of 8mm-mrad was used. The diameter of laser spot was 0.32mm focused by a focus lens with a focal length of 300mm. The motion system was a six-axis produced by Arnold Company. The laser was applied vertically to the upper surface of specimen during the welding process. After irradiating the melt pool with a low-power laser at a wavelength of 808nm, a PHOTRON Fastcam Mini UX100 color high-speed camera was used to photograph the keyhole aperture and the molten pool surface at a frame rate of 1.6×10^4 frames/s. The high-speed camera was arranged on the side perpendicular to welding speed, and the angle between its axis and laser axis was about 10° . The experimental setup is shown in Fig. 3(a).



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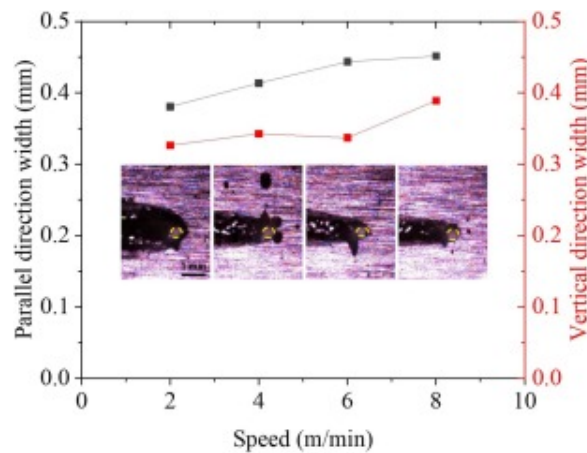
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Fig. 3. Schematic diagram of the experimental setup.

While shooting by using the high speed camera, low carbon steel was chosen as the processing material. The chemical composition of the low-carbon steel is 0.12wt% C, 0.26wt% Si, 0.33wt% Mn, 0.019wt% P, 0.018wt% S and the balance of Fe. When switching to copper hafnium alloy as welding material, as shown in Fig. 3(b), the laser was suddenly turned off during scanning process and the molten pool was rapidly frozen to preserve the actual keyhole morphology. The total scanning time of laser (i.e., t_2 in Fig. 3) was 2 s, the output power time of the 5 kW laser (i.e., t_1 in Fig. 3) was set as 0.2 s, and the laser's output power was 0 kW in the rest time. In addition, the specimens were wiped clean with acetone before the welding experiments to avoid the influence of impurities on welding experiments. The dimensions of both materials were 100 mm $\times$ 50 mm $\times$ 10 mm.

3.2. Experimental validation

The laser power was fixed at 4 kW and the defocus amount was 0 mm. The morphology and the width of keyhole aperture with the variation of welding speed based on the observation by high speed camera were obtained, as shown in Fig. 4. It can be seen that the keyhole aperture width in the direction paralleling with welding speed was in the range of about 0.37–0.46 mm; and the keyhole aperture width in the direction vertical with welding speed was relatively small about 0.3–0.39 mm in the range. The keyhole aperture width and the spot diameter obtained by experimental measurements were basically in the same order of magnitude within the corresponding parameters, that is, the diameter of keyhole aperture is slightly larger than the diameter of laser spot. The morphology of keyhole aperture cannot change significantly after increasing the welding speed.



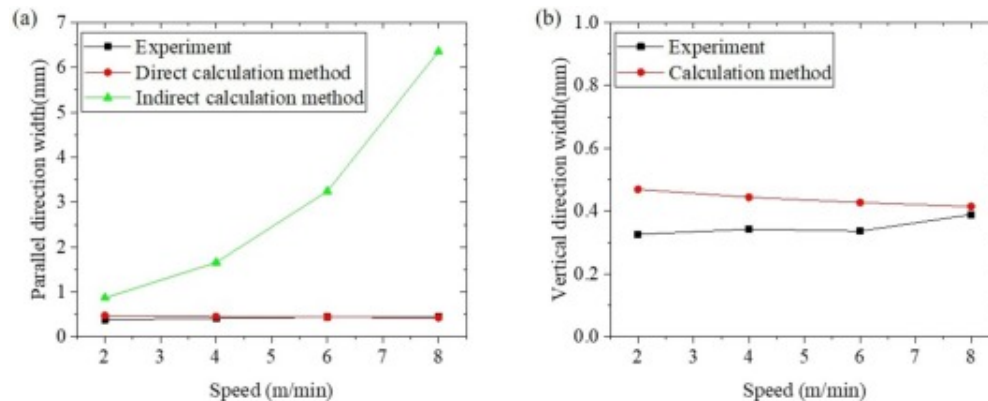
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Fig. 4. The morphology of keyhole aperture and molten pool surface and the measurement results about the keyhole aperture width.

Based on the above two algorithms, the relationship between the width of keyhole aperture in the direction paralleling with welding speed from calculation and experimental measurement and the welding speed was obtained as shown in Fig. 5 (a). The width of keyhole aperture calculated by the indirect calculation method increased rapidly with the increase of welding speed. When the welding speed was high, the calculated keyhole aperture width in the direction parallel with welding speed was more than 5 mm, much larger than the experimental measurement results. On the contrary, the results calculated by the direct calculation method were the same as the experimental measurement results. The results about the width of keyhole aperture in the direction vertical to welding speed calculated by using the direct calculation method are shown in Fig. 5(b). It can be seen that the calculated results and experimental measurements remained nearly 0.4 mm (slightly larger than the diameter of laser spot). The difference between calculated results and experimental measurements was small. By comparing the calculated and experimental measurements of the two algorithms, it can be concluded

that the direct calculation method is more reasonable for estimating the keyhole aperture profile. Therefore, this method was used for all calculations in the following. In addition, the high-speed camera was not coaxial with laser, so it will affect the measurement results obtained from captured images. During the experiments, the high-speed camera was arranged on the side perpendicular to welding speed, and the angle between its axis and laser axis was about 10° . The angle will not affect the keyhole aperture width in the direction parallel to welding speed, which made the results of experimental and direct calculation methods in Fig. 5(a) more consistent. However, the tilt of shooting angle will cause the keyhole aperture width in the direction vertical to welding speed measured from images to be smaller than the actual width, which will make the experimental measurement results smaller than the calculated results (Fig. 5(b)).



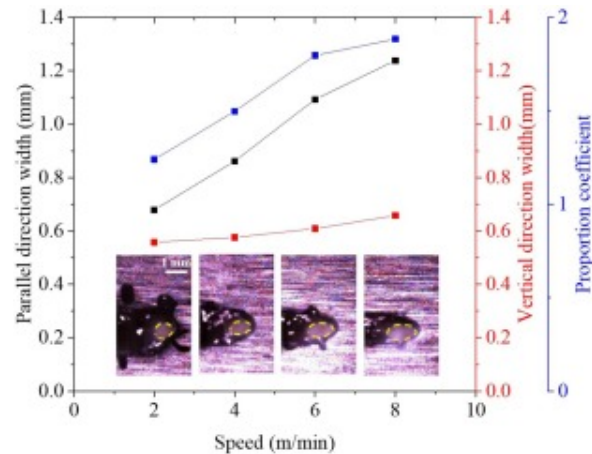
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Fig. 5. The relationship between welding speed and keyhole aperture width in the direction paralleling (a) and vertical (b) with welding speed from calculations and experimental measurements.

4. The effect of laser-induced vapor erupted from front keyhole wall on the keyhole aperture morphology

During the laser deep penetration welding, changing the amount of evaporation from laser-induced keyhole wall or the tilt angle of keyhole wall (the direction of vapor emission) can change the impact effect of laser-induced vapor on keyhole wall (affecting the morphology of keyhole aperture) [24]. The variation of welding speed can result in a significant change in the tilt angle of front keyhole wall. The laser power was fixed at 4kW and the defocused amount was +10mm. Changing the welding speed about 2, 4, 6 and 8m/min, the relationship between the measurement results about keyhole aperture morphology and the keyhole aperture width in the direction parallel and perpendicular to welding speed and its scale factor with the variation of welding speed was obtained as shown in Fig. 6. It can be seen that as the welding speed increased, the morphology of keyhole aperture gradually changed from nearly circular to oval. The width of keyhole aperture in the direction parallel and vertical to welding speed was subsequently increased, but the ratio coefficient of the length and width of keyhole aperture was different, that is, the morphology of keyhole aperture evolved obviously from the approximate circle to oval.

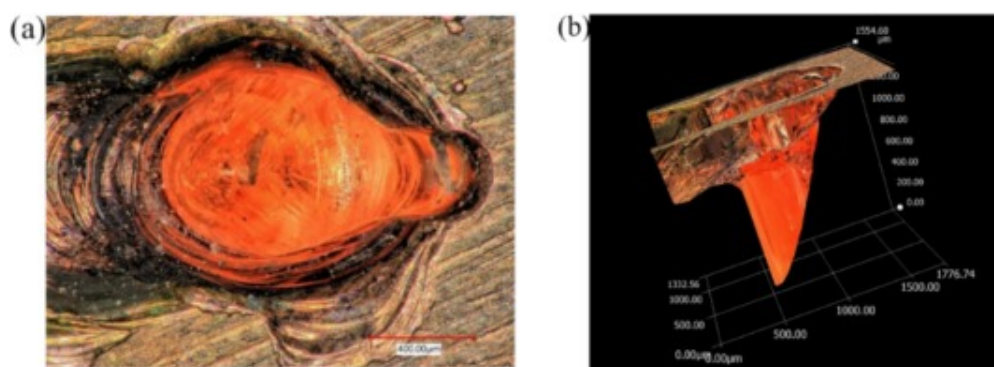


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Fig. 6. The relationship between the morphology and geometrical characteristic of keyhole aperture and welding speed.

The absorptivity of low carbon steel to fiber laser is high, so the keyhole depth is large, the tilt angle of front keyhole wall is relatively small, and the impact of vapor eruption on molten pool around the keyhole wall is not obvious. Moreover, the characteristics of the copper and hafnium alloy about the low absorptivity of incident fiber laser (shallow weld depth, relatively small tilt angle of front keyhole wall), low-temperature difference between the melting and boiling temperature and the high thermal conductivity are considered. Consequently, the laser is suddenly turned off during the welding process of copper and hafnium alloy, the melt pool quickly freezes and the keyhole will be retained to get the morphology of keyhole aperture. During the experiment, the laser power was fixed at 5kW, the welding speed was 2m/min, the defocusing amount was 0mm and the radius of focused laser spot was approximately 0.125mm. The morphology of keyhole aperture and the three-dimensional morphology of keyhole can be obtained as shown in Fig. 7.



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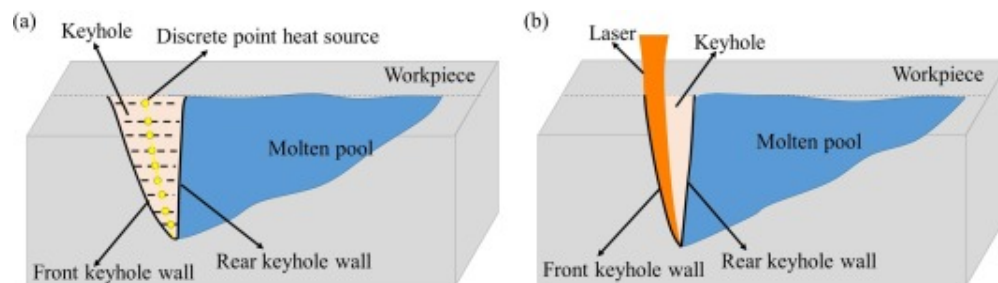
Fig. 7. The morphology of keyhole aperture (a) and the three-dimensional morphology of keyhole (b) during laser deep penetration welding of copper-hafnium alloy.

As can be seen in Fig. 7(a), the keyhole aperture was divided into a semi-circular region with a small area at the front end (formed by direct laser heating) and a circular region with a relatively large area at the rear end (formed by laser-induced vapor emission from

front wall within the keyhole). Measuring the width of the two regions separately, it can be seen that the half-width of the long axis of semi-elliptical region at the front end of keyhole aperture was approximately 0.149mm and the half-width of the short-axis was approximately 0.16mm, which is slightly larger than the radius of focused laser spot. The radius of nearly circular area at the rear end was approximately 1 mm, which is much larger than the radius of focused laser spot. The corresponding keyhole depth (approximately 1.6mm), the keyhole aperture width (approximately 1.3mm) and the inclination angle of keyhole front wall (approximately 50°) were also measured from the 3D keyhole morphology. It can be seen that when the inclination angle of front keyhole wall was relatively small, the vapor eruption region of keyhole aperture was obvious.

5. Analysis and discussion

Considering that the keyhole has been formed in melt pool, the laser energy in keyhole was discretized in different depth planes to form a collection of point heat sources on different depth planes to estimate the temperature field of each point heat source on plate surface (indirect calculation method), which is most consistent with the characteristics of laser deep penetration welding, as shown in Fig. 8(a). However, we found in the previous study [25], [26] that the laser radiated on front keyhole wall in fiber laser deep penetration welding, as shown in Fig. 8(b). It can be indicated that the rear keyhole wall was barely irradiated by incident laser, and the laser energy absorbed by front keyhole wall needs to be thermally transferred around the keyhole area to the rear keyhole wall and the end of keyhole. Therefore, the heat source for calculating keyhole morphology was constructed in Fig. 8(a) resulting in a discrete heat source with a greater intensity than that used in actual welding. It may be the reason why the keyhole aperture profile calculated by indirect calculation method was larger than the it observed.



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Fig. 8. Schematic representation of the action between laser and keyhole in the indirect calculation method (a) and in the actual welding process (b).

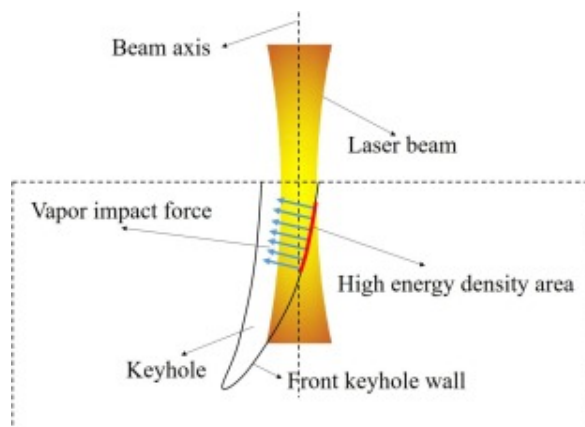
At the beginning of laser deep penetration welding, laser beam interacts with workpiece surface and the material absorbs laser energy up to its evaporation threshold and vaporizes rapidly. At this time, it can be assumed that the intensity of heat source acting on material surface remains consistent with the laser energy distribution. After the vaporization of material on workpiece surface, the evaporative recoil pressure acts downwards and forms the keyhole. When ignoring the effect of laser-induced vapor on keyhole morphology, it can be assumed that the keyhole aperture portion where the material surface has vaporized at the beginning of welding is similar to the area directly heated by the laser. When the keyhole was formed, the laser energy was absorbed by the front and rear keyhole wall unsymmetrically, as shown in Fig. 8(b). The region of evaporation temperature on molten pool surface, assuming that the laser heat source was a laser-induced evaporation heat source on a thin planar layer (direct calculation method), was consistent with the in-situ observation of the keyhole aperture, which indicated that the energy distribution of laser beam on front keyhole wall can be ignored while calculating the morphology of the direct laser action region of keyhole aperture, and the direct calculation method is reasonable.

The difference between the calculated keyhole aperture profile and the experimental measurements was more pronounced when the evaporative vapor effect on keyhole wall was evident, as shown in Fig. 6, Fig. 7. The above methods for estimating keyhole aperture profile were based on the law of heat conduction, considering the absorbed laser energy and the energy required to heat material from ambient temperature to boiling temperature. However, in laser deep penetration welding, the laser acts directly on front keyhole wall. The laser-induced evaporative vapor from the front keyhole wall erupts mainly on the rear keyhole wall, and the melt near the keyhole aperture will be expanded outward under the force of vapor eruption. Therefore, the keyhole aperture can be divided into a region formed by direct laser heating and the other region formed by laser-induced vapor emission from the keyhole wall. The above estimation cannot obtain the region formed by the eruption of laser-induced vapor on keyhole wall. Therefore, there were differences between calculation results and experimental results under different parameter conditions.

The directions of the recoil pressure for evaporating vapor on front keyhole wall and the impact force on rear wall are exactly opposite. The magnitude of recoil pressure and impact force is related to the beam intensity absorbed at various points on front wall, which can be approximated as [27]:

$$P_r = K \bullet I_{abs} \quad (11)$$

where I_{abs} represents the absorbed laser intensity. The constant K is related to material and process parameters. According to Fig. 9, the eruption of laser-induced vapor on front keyhole wall will impact on the molten pool nearly the rear wall and the tail of keyhole aperture, which will have an impact on keyhole aperture morphology.

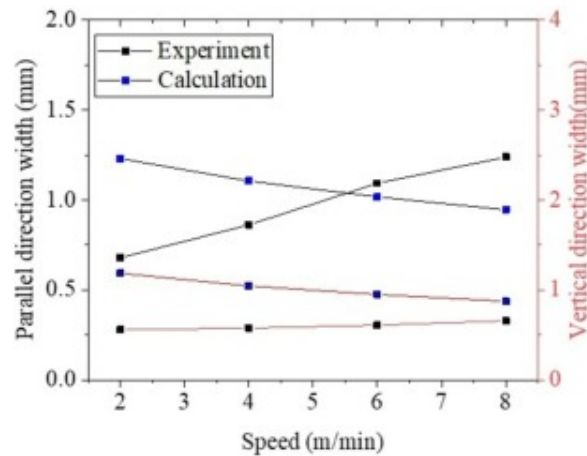


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Fig. 9. Schematic diagram of the interaction between laser and keyhole.

During the welding process, the change of welding speed significantly altered the inclination angle of front keyhole wall, thus changing the impact force direction of vapor eruption striking on rear keyhole wall. Based on the direct calculation method and the experimental results in Fig. 6, the relationship between calculated and experimental measurements of keyhole aperture width in the direction of parallel and perpendicular to welding speed under different welding speed, and the results are shown in Fig. 10. It can be seen that the calculated keyhole aperture width decreased when the welding speed increased, while the experimental measurement results were exactly the opposite. For the keyhole aperture width in the direction parallel to welding speed, when the welding speed was low, the calculated results were relatively large; when the welding speed was high, the experimental measurement results were greater than the calculated results. At the same time, the difference gradually increased with the increase of welding speed.



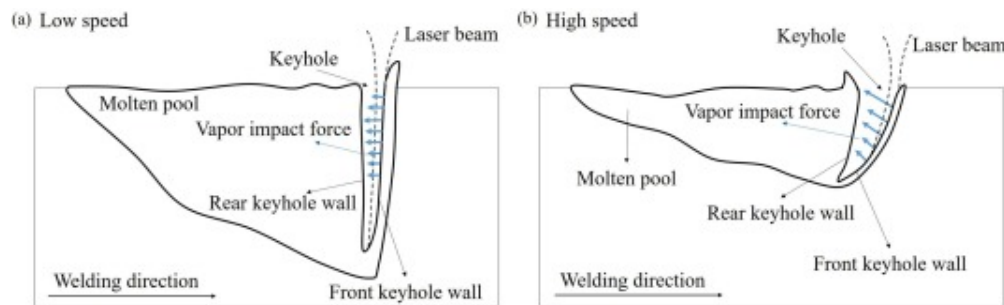
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Fig. 10. The relationship between the calculated keyhole width and the experimental measurement results in parallel and vertical to welding speed directions under different welding speed.

For the keyhole cross section in the direction parallel to welding speed, when the welding speed was small, the laser energy input was high, as shown in Fig. 11(a). The inclination angle of front keyhole wall was small and approximately closed to vertical. As a result, the direction of impact force on the rear wall caused by laser-induced evaporating vapor was approximately parallel to welding speed. Consequently, the evaporative vapor from front wall nearly keyhole top mainly acted on the rear and end of keyhole aperture to push melt pool flow. In addition, the recoil pressure of evaporating vapor on front wall and the eruption impact force will form a raised liquid column in the molten pool at the front and rear of keyhole aperture [28], which affects the actual measurement of keyhole aperture, obtaining a slight reduction about the keyhole aperture width. Therefore, when the welding speed was low, the calculated keyhole aperture width in the parallel direction to welding speed will be slightly greater than the experimental measurement. When the welding speed was high, as shown in Fig. 11(b), the inclination angle of front keyhole wall

became larger. Compared to low welding speed, the range of evaporated vapor from front wall acted on the rear and end of keyhole aperture was increased. Therefore, as the welding speed increased, the influence of evaporative vapor on keyhole aperture width in the direction parallel to welding speed increased. Therefore, changing the inclination angle of front wall caused a change in the direction of vapor action, causing the keyhole aperture morphology to evolve from circular to oval, as shown in Fig. 6.



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Fig. 11. Schematic diagram of laser deep penetration welding with low speed (a) and high speed (b).

In contrast to Fig. 4 and Fig. 6, when the defocus amount was 0mm, the change of keyhole aperture morphology with welding speed was not obvious, while the effect of vapor on the keyhole aperture morphology and the two parts of keyhole aperture cannot be reflected. The low-carbon steel material has a high absorptivity of fiber laser. The laser energy absorbed by the material remained high at the defocus amount of 0mm, even when the welding speed was nearly 8m/min. Therefore, the keyhole depth was relatively deep, the inclination angle of front keyhole wall and the variation with increasing welding speed were small. The change of action position of evaporating vapor was not sufficient to change the eruption trend of melt at the rear and end of keyhole aperture. Accordingly, the change law of keyhole aperture morphology with welding speed was not obvious. While welding copper and hafnium alloys, the absorptivity to fiber laser was

much smaller than that of low-carbon steel, and the keyhole depth was also relatively smaller, as shown in Fig. 7. Therefore, the inclination angle of front keyhole wall was larger, and the effect of metal evaporating vapor from front wall on the keyhole aperture was obvious. The morphology of keyhole aperture obtained was oval, and the distinction between the two parts of keyhole aperture was also more obvious.

In summary, the keyhole aperture during laser deep penetration welding can be divided into the zone formed by laser direct heating and the zone formed by laser-induced vapor eruption from front keyhole wall. When the welding speed was low or the keyhole depth was large, the zone formed by laser-induced vapor eruption from front keyhole wall was not obvious, and the zone formed by laser direct heating was similar to a circle. Changing the welding speed or material will make the inclination angle of front keyhole wall smaller, and the impact effect of laser-induced vapor from the front wall on the rear wall was then increased, resulting in the keyhole aperture morphology becoming elliptical or “gourd” and forming the zone formed by laser-induced vapor eruption from the front keyhole wall. Assuming that the melt temperature at the edge of keyhole aperture was boiling temperature, the boiling temperature profile of the laser-induced molten pool surface on material surface calculated by the direct calculation method can be used to calculate the zone formed by laser direct heating approximately.

6. Conclusion

In this paper, it was assumed that the temperature of molten liquid at the edge of keyhole aperture is boiling point temperature, and the boiling temperature isotherm on melt pool surface was calculated based on two different treatment methods of heat source to estimate keyhole aperture profile, which was verified then. The following conclusions can be obtained:

- (1) The keyhole aperture in laser deep penetration welding can be divided into the zone formed by laser direct heating and the zone formed by laser-induced vapor eruption from the front keyhole wall. The former is closely

related to the laser energy distribution on material surface rather than the laser energy distribution in the direction of keyhole depth, while the latter is more susceptible to the influence of erupted laser-induced vapor related to the energy distribution on front keyhole wall. When the inclination angle of front keyhole wall was large, the zone formed by laser-induced vapor eruption from the front keyhole wall was not obvious and the zone formed by laser direct heating was similar to a circle.

- (2) Assuming the contour of boiling temperature on molten pool surface to be the keyhole aperture contour, and the zone formed by laser direct heating can be approximated obtained by estimating the boiling temperature isotherm (characterizing the keyhole aperture contour) on material surface according to heat conduction law by using the laser heat source acting on molten pool surface.
- (3) When the inclination angle of front keyhole wall became smaller by changing the welding speed or welding materials and other factors, the impact effect on the rear keyhole wall will be enhanced when the laser-induced vapor erupted on the surface of front keyhole wall, resulting in the formation of the zone formed by laser-induced vapor eruption from the front keyhole wall gradually obvious and the keyhole aperture morphology evolving into an ellipse or “gourd” morphology.

CRediT authorship contribution statement

Baoqi Zhu: Writing – original draft, Project administration, Methodology, Formal analysis, Data curation. **Wuxiong Yang:** Writing – review & editing, Validation, Data curation, Conceptualization. **Jianglin Zou:** Writing – review & editing, Resources, Project administration, Funding acquisition, Data curation, Conceptualization. **Shun Xie:**

Supervision, Methodology, Investigation. **Chao Fang:** Writing – review & editing, Software, Conceptualization. **Tao Liu:** Validation, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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